# ENEL 585 – Introduction to Power Electronics

Fall 2026 · Quiz 1 · Tuesday, September 29, 2026 · Worked solutions

Instructor: Prof. Philippe Gray

This is the quiz as written, with a worked solution in the box under each part. The quiz has **3 questions** worth **23 marks**; it is graded **out of 19**, with raw marks unchanged.

**Conditions that applied to every part**

- Unless a part says otherwise: all components are ideal, the converter is in steady state, and the output voltage ripple is small enough to treat $v(t) \approx V$.
- **Show your working.** A correct final number with no working shown earned at most half the marks for that part. One line is enough, for example the balance equation you set up or the value you read from a waveform.

**Notation** (as in the lectures): $v_g$ / $V_g$ input voltage (instantaneous / dc), $v$ / $V$ output voltage (instantaneous / dc), $D$ duty ratio of the controlled switch, $D' = 1 - D$, $T_s = 1/f_s$, $i_L$ / $I_L$ inductor current (instantaneous / average), $\Delta i_L$ peak-to-peak inductor current ripple, $\Delta v$ peak-to-peak output voltage ripple.

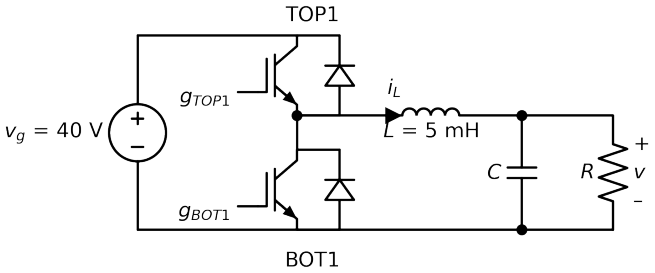
**The two tools used throughout**

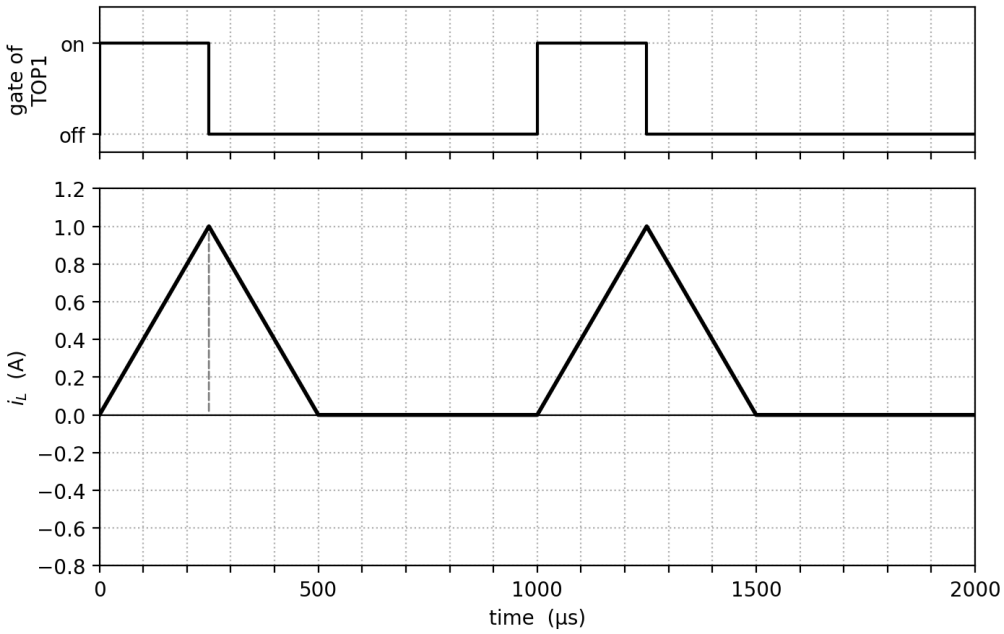
- **Volt-second balance:** in steady state the average inductor voltage over one period is zero, $\langle v_L \rangle = 0$.
- **Charge balance:** in steady state the average capacitor current over one period is zero, $\langle i_C \rangle = 0$.

Every answer below comes from one of these two statements applied to the intervals of the switching period. No converter-specific formula is needed.

## Question 1 – Reading a bench waveform (10 marks)

The Lab 1 buck converter is operated open-loop with $v_g = 40$ V, $L = 5$ mH, $f_s = 1$ kHz. TOP1 is gated with $D = 0.25$. **BOT1 is not gated**, so only its antiparallel diode can conduct. The oscilloscope capture below shows the TOP1 gate signal and the inductor current $i_L$.




**(a)** [2] In which conduction mode is the converter operating? Name the feature of the trace that establishes it, and the device property that causes it.

Solution

**Discontinuous conduction mode (DCM).**

- *Feature of the trace:* $i_L$ stays at zero from 500 $\mu$s to 1000 $\mu$s, half of every period.
- *Device property:* with BOT1 not gated, only its antiparallel diode can carry the freewheeling current, and a diode cannot conduct in reverse. When $i_L$ reaches zero the diode turns off and the current stays at zero instead of going negative.

"Light load" explains why DCM occurs in general, but it is not the device property the question asks for.

**(b)** [3] Determine the average output voltage $V$ from the trace. Show your reasoning.

Solution

Read the three intervals of one period ($T_s = 1000\ \mu$s) from the trace:

| Interval | Duration | Conducting | $v_L$ |
|---|---|---|---|
| 0 to 250 $\mu$s | $DT_s$, with $D = 0.25$ | TOP1 | $v_g - V$ |
| 250 to 500 $\mu$s | $D_2T_s$, with $D_2 = 0.25$ | BOT1 diode | $-V$ |
| 500 to 1000 $\mu$s | $D_3T_s$, with $D_3 = 0.5$ | nothing | 0 (because $i_L$ is constant at zero) |

Volt-second balance over the full period:

$$(v_g - V)\,D + (-V)\,D_2 + 0 \cdot D_3 = 0 \quad \Rightarrow \quad V = v_g\,\frac{D}{D + D_2} = 40 \times \frac{0.25}{0.25 + 0.25} = \mathbf{20\ V}$$

*Second route, from the ripple:* the trace gives $\Delta i_L = 1.0$ A, built up while TOP1 is on. With $v_L = v_g - V$ in that interval, $\Delta i_L = \dfrac{1}{L}\displaystyle\int_0^{DT_s} v_L\,dt = \frac{(v_g - V)\,DT_s}{L}$, so $1.0 = \dfrac{(40 - V)(250\ \mu\text{s})}{5\ \text{mH}}$ and $V = 20$ V.

*Pitfall:* $V = Dv_g = 10$ V. That result assumes $v_L = -V$ for the whole off-time $D'T_s = 750\ \mu$s, which is true only in CCM. Here the diode conducts for only 250 $\mu$s, so less negative volt-seconds are applied and $V$ has to be higher for the balance to hold.

**(c)** [2] Determine the load resistance $R$ that was connected during this measurement.

Solution

Charge balance on $C$: the average capacitor current is zero, so in a buck converter the average inductor current equals the load current, $I_L = V/R$.
The average of $i_L$ is the area of the triangle divided by the **full** period:

$$I_L = \frac{1}{T_s} \cdot \frac{1}{2}\,(1.0\ \text{A})(500\ \mu\text{s}) = \frac{250\ \mu\text{A} \cdot \text{s}}{1000\ \mu\text{s}} = 0.25\ \text{A}$$

$$R = \frac{V}{I_L} = \frac{20\ \text{V}}{0.25\ \text{A}} = \mathbf{80\ \Omega}$$

**(d)** [3] BOT1 is now gated **complementarily** to TOP1 (synchronous buck). $D = 0.25$ and $R$ are unchanged. On the blank axes below, sketch $i_L(t)$ over two switching periods, labelling the peak, valley and average values. State the new output voltage $V$.

Solution

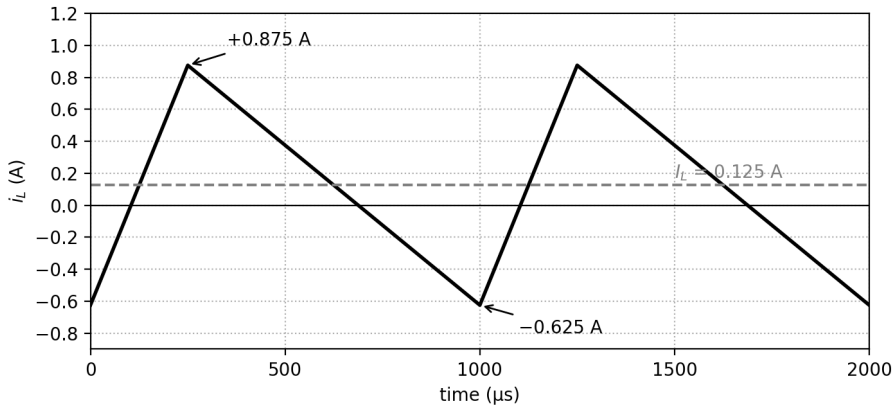
With BOT1 gated, the lower switch position conducts in both directions: its diode carries positive $i_L$ and its IGBT carries negative $i_L$. Nothing stops the current at zero any more, so $v_L = -V$ for the whole off-interval and the converter operates in **CCM at any load**.

*Output voltage:* $(v_g - V)D - VD' = 0 \Rightarrow V = Dv_g = 0.25 \times 40 = \mathbf{10\ V}$. The output voltage halves even though $D$ and $R$ did not change.

*Average:* $I_L = V/R = 10/80 = 0.125$ A.

*Ripple:* $\Delta i_L = \dfrac{(v_g - V)\,DT_s}{L} = \dfrac{30\text{ V} \times 250\ \mu\text{s}}{5\text{ mH}} = 1.5$ A.

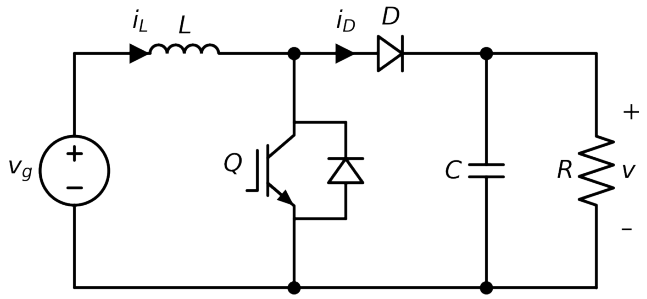
*Peak and valley:* $I_L \pm \Delta i_L/2$, so the peak is $0.125 + 0.75 = +\mathbf{0.875\ A}$ at $t = 250\ \mu$s and the valley is $0.125 - 0.75 = -\mathbf{0.625\ A}$ at $t = 0$ and $1000\ \mu$s.




The ripple (1.5 A) is much larger than the average (0.125 A), so the current is negative for part of every period. While it is negative it flows through the BOT1 IGBT during the off-interval and through TOP1's antiparallel diode at the start of the on-interval.

## Question 2 – Boost converter (5 marks)

The Lab 1 boost converter below is supplied from $v_g = 15$ V with $L = 5$ mH and $f_s = 5$ kHz. The controller adjusts $D$ so that the output is regulated at $V = 45$ V. The load is $R = 25\ \Omega$ (ten load-bank switches closed). Assume CCM.




**(a)** [3] Sketch $i_L(t)$ over one switching period. Label, with numerical values: the slope in each interval, the average value $I_L$, and the peak-to-peak ripple $\Delta i_L$. State the average diode current and how you obtained it.

Solution

*Duty ratio.* Volt-second balance for the boost: $v_g D + (v_g - V)D' = 0 \Rightarrow V = v_g/D'$, so

$$D' = \frac{v_g}{V} = \frac{15}{45} = \frac{1}{3}, \qquad D = \frac{2}{3}$$

With $T_s = 1/f_s = 200\ \mu$s: $DT_s = 133.3\ \mu$s and $D'T_s = 66.7\ \mu$s.
*Ripple.* While the switch is on, $v_L = v_g$:

$$\Delta i_L = \frac{1}{L}\int_0^{DT_s} v_L\, dt = \frac{v_g\, DT_s}{L} = \frac{15\text{ V} \times 133.3\ \mu\text{s}}{5\text{ mH}} = \mathbf{0.4\ A}$$

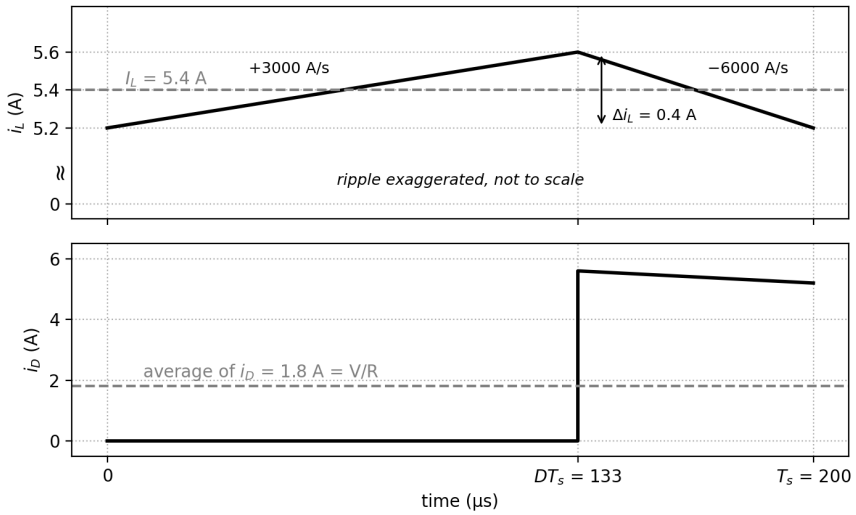
*Slopes* (asked for on the sketch), from $\frac{di_L}{dt} = \frac{v_L}{L}$:

- Switch on (0 to 133.3 $\mu$s): $v_L = v_g = 15$ V, so $\frac{di_L}{dt} = \frac{15}{5\text{ mH}} = +\mathbf{3000\ A/s}$.
- Switch off (133.3 to 200 $\mu$s): $v_L = v_g - V = -30$ V, so $\frac{di_L}{dt} = \frac{-30}{5\text{ mH}} = -\mathbf{6000\ A/s}$.

*Average diode current.* Charge balance on $C$: the diode is the only path that feeds the output node, so $\langle i_D \rangle = I = V/R = 45/25 = \mathbf{1.8\ A}$.
*Average inductor current.* The diode carries $i_L$ only during the off-interval, so $\langle i_D \rangle = I_L D'$:

$$I_L = \frac{I}{D'} = \frac{1.8}{1/3} = \mathbf{5.4\ A}$$

Power balance gives the same result: $\frac{V^2}{R} = v_g I_L \Rightarrow I_L = \frac{V^2}{R\, v_g} = \frac{45^2}{25 \times 15} = 5.4$ A.
$i_L$ therefore swings between $I_L - \Delta i_L/2 = 5.2$ A and $I_L + \Delta i_L/2 = 5.6$ A. It rises for two thirds of the period and falls for the last third. The sketch exaggerates the ripple so that the shape is visible.



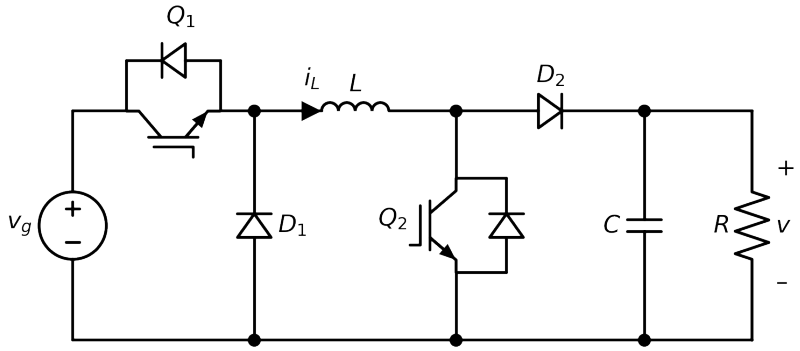

**(b)** [2] The load bank is switched to a single 250 Ω resistor. The controller still holds $V = 45$ V. Does the converter remain in CCM? Justify with numbers.

Solution

As in the lecture notes, the converter is in DCM when $I_L < \Delta i_L/2$ and in CCM when $I_L > \Delta i_L/2$.
*Ripple.* $\Delta i_L = v_g DT_s/L$ is not load dependent. Since $v_g$ and $V$ are unchanged, $D$ is unchanged, so the ripple is still 0.4 A and $\Delta i_L/2 = 0.2$ A.
*Average.* $I = V/R = 45/250 = 0.18$ A, so $I_L = I/D' = 0.18 \times 3 = 0.54$ A.
*Comparison.* $I_L = 0.54$ A $> 0.2$ A, so the valley current is $0.54 - 0.2 = +0.34$ A. **The converter remains in CCM.**
*Where the boundary is:* $I_L = 0.2$ A corresponds to $I = 0.2 \times D' = 0.067$ A, that is $R = 45/0.067 = 675$ Ω.
*Pitfall:* comparing the **load** current $I = 0.18$ A with 0.2 A and concluding DCM. The comparison has to use the inductor current, which in a boost is $1/D'$ times the load current (three times, here).

## Question 3 – An unfamiliar converter (8 marks)

The converter below uses two IGBTs $Q_1$ and $Q_2$, each with an antiparallel diode as on the lab hardware, and two diodes $D_1$ and $D_2$. $Q_1$ and $Q_2$ receive the **same** gate signal: both are on for $0 \le t < DT_s$ and both are off for $DT_s \le t < T_s$. The converter operates in CCM.




**(a)** [2] For each of the two intervals, state which of $Q_1$, $Q_2$, $D_1$, $D_2$ conduct and write $v_L(t)$ and $i_C(t)$ in terms of $v_g$, $V$, $i_L$ and $R$.

Solution

*Interval 1, $0 \le t < DT_s$ ($Q_1$ and $Q_2$ on).* $Q_1$ connects the left end of $L$ to $v_g$ and $Q_2$ connects the right end to ground. $D_1$ has $v_g$ across it in reverse and $D_2$ has $V$ across it in reverse, so both are off. The current path is $v_g \to Q_1 \to L \to Q_2 \to$ ground, and the capacitor alone supplies the load.

$$v_L = v_g, \qquad i_C = -\frac{V}{R}$$

*Interval 2, $DT_s \le t < T_s$ ($Q_1$ and $Q_2$ off).* The inductor current cannot stop, so it forces $D_1$ and $D_2$ on. $D_1$ connects the left end of $L$ to ground and $D_2$ connects the right end to the output. The current path is ground $\to D_1 \to L \to D_2 \to$ output.

$$v_L = -V, \qquad i_C = i_L - \frac{V}{R}$$

The antiparallel diodes of $Q_1$ and $Q_2$ never conduct: each is reverse-biased whenever its IGBT is off, as part (d) shows.

**(b)** [2] Derive the conversion ratio $M(D) = V/V_g$.

Solution

Volt-second balance with the two expressions for $v_L$ from (a):

$$v_g D + (-V)D' = 0 \quad \Rightarrow \quad M(D) = \frac{V}{V_g} = \frac{D}{1-D}$$

$M < 1$ for $D < 0.5$ and $M > 1$ for $D > 0.5$, so the converter steps both down and up. The output is positive, as the $+$ and $-$ on the figure show: this is a **non-inverting** buck-boost.

**(c)** [2] Derive the average inductor current $I_L$ in terms of $V$, $R$ and $D$. Then express it in terms of the load current $I = V/R$ and $D$.

Solution

Charge balance with the two expressions for $i_C$ from (a), using $i_L \approx I_L$:

$$\left(-\frac{V}{R}\right) D + \left(I_L - \frac{V}{R}\right) D' = 0 \quad \Rightarrow \quad I_L D' = \frac{V}{R}$$

$$I_L = \frac{V}{R\,(1-D)} = \frac{I}{1-D}$$

The inductor feeds the output only during $D'T_s$, so its average has to be larger than the load current by the factor $1/D'$. This is the same relation as in the boost converter of Question 2.

**(d)** [2] What voltage must $Q_1$ block when it is off? What voltage must $Q_2$ block? Compare this converter with another converter covered in the course that can also both step up and step down the voltage: name the converter you are comparing against, and give one advantage and one disadvantage of the converter above.

Solution

*Blocking voltages.* Look at each IGBT during interval 2, when it is off.

- $Q_1$: its collector is at $v_g$ and its emitter is held at ground by the conducting $D_1$. $Q_1$ **blocks $v_g$.**
- $Q_2$: its emitter is at ground and its collector is held at $V$ by the conducting $D_2$. $Q_2$ **blocks $V$.**

*Comparison.* The comparison has to be against a converter that steps both up and down, so a buck or a boost alone does not qualify. Two examples:

| Compared against | Advantage of the converter above | Disadvantage of the converter above |
|---|---|---|
| Inverting buck-boost (one switch, which blocks $v_g + \lvert V\rvert$; output inverted) | Each device blocks only $v_g$ or $V$, so lower-voltage parts can be used. The output is not inverted. | Four semiconductors instead of two. Two devices are in series in every current path, so the conduction loss is about double. $Q_1$ needs a floating gate drive. |
| Flyback (switch blocks $v_g + (N_1/N_2)V$ plus a leakage spike) | Lower device voltage. No transformer to design and no leakage-inductance spike to snub. | No galvanic isolation, and no turns ratio to reach a large step-up or step-down. Four semiconductors, with two in series in every current path. |

"Non-inverting" is not an advantage over the flyback, because the flyback output polarity is set by how the secondary winding is connected.